Network-Level Insights into Google-Free Android Operating Systems

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Abstract. This paper presents a comparative analysis of user privacy in Android operating systems through network traffic inspection. Using a controlled man-in-the-middle setup and traffic decryption techniques, six Android systems were evaluated: stock Android (GMS), GrapheneOS, iodéOS and 3 variants of LineageOS (stock, Gapps, microG). While prior work focuses on applications or stock Android across device manufacturers, this study investigates alternative Android distributions designed with privacy as a primary objective. We examined outbound connections, transmitted identifiers, and network traffic patterns to assess the impact of integrated Google services on user privacy. Our findings demonstrate that stock Android maintains the most communication with Google servers, transmitting identifiers and metadata even while not actively used. In contrast, all privacy-focused systems minimize unsolicited traffic and offer stronger user control. These results provide practical insights into how alternative Android distributions can improve user privacy without sacrificing core functionality, highlighting the potential of open-source ecosystems as viable, privacy-respecting alternatives to Google-dependent mobile environments.

Keywords: Android · Privacy · Network traffic.

1 Introduction

Android is the dominant mobile operating system and acts as a primary gateway to digital world for billions of users. It is crucial for core operating system to be a reliable, stable and trustworthy platform, that allows users to run applications and services of their choice. While third-party application privacy has been widely studied [2, 10, 8, 13], less attention is typically paid to the operating system layer and to privileged service frameworks that silently enable additional functionality. Standard unprivileged applications are subject to Android permission model, although some attempt to circumvent these restrictions in creative ways [9, 7]. In contrast, privileged system applications operate with higher access to the device and OS, which makes their actions harder to understand and control. Most user-facing privacy controls focus on applications and permissions, yet a substantial amount of data exchange can occur below, through

OS services, vendor components and integrated Google components. This work therefore compares the network behavior of multiple Android distributions and their preinstalled/privileged system components under controlled, reproducible conditions.

1.1 Android dependency on privacy-threatening services

While Android core is the Android Open Source Project (AOSP), most consumer devices include additional proprietary components layered on top. They are often implemented as privileged system applications and services to provide baseline functionality such as push notifications, location assistance, application distribution/updates, device integrity checks and many more. Users and application developers commonly assume their presence and availability by default.

These components exists for a practical reason. Centralized services can significantly improve usability and efficiency, for example by reducing battery drain compared to per-application background connections, and by providing faster, more accurate location through fused network and sensor-based methods. However, this design also tightly couples many functions into a single service stack. Enabling useful capabilities may implicitly enable others that a user would not choose, such as advertising identifiers, telemetry, and behavioral tracking. In practice, the result is a large proprietary bundle that is difficult to audit and decompose into only the features a user actually wants.

On Android, these baseline capabilities are most commonly delivered through GMS (Google Mobile Services). Because Google's business model is strongly advertising-driven, many users may be unwilling to entrust this provider with broad access to their devices and data. In principle, users should be able to choose which services they rely on and have confidence that these choices are enforced by the platform's permission and isolation mechanisms. This demand for transparency and controllability motivates alternative approaches such as microG[1] and GrapheneOS sandboxed Google Play[2]. These solutions aim to preserve device functionality and application compatibility while reducing privilege and improving user control over Google-related components.

1.2 Privacy-focused Android operating systems

Privacy-focused Android operating systems are distributions derived from AOSP that aim to reduce data exposure at the OS level. They achieve this by limiting preinstalled privileged components, tightening default settings, and providing more transparent controls over permissions and network access. In contrast to stock Android (the factory system shipping with Google Mobile Services enabled by default), privacy-oriented systems minimize background communication, restrict persistent identifiers, and strengthen user enforcement mechanisms over non-essential components.

[1] `https://github.com/microg/GmsCore/wiki`
[2] `https://grapheneos.org/usage#sandboxed-google-play`

In this work we focus on three representative approaches: GrapheneOS, iodéOS and 3 variants of LineageOS (stock, Gapps, microG). GrapheneOS[3] prioritizes a hardened security model and strong isolation. Google components are not included by default, but can be installed in a dedicated sandboxed mode as ordinary applications, without the special privileges they typically hold on factory systems. This design targets improved containment and user control while preserving high compatibility when user decides to install Google services. GrapheneOS also improves low-level security, for example by replacing Android's default memory allocator with its own hardened implementation to make memory corruption exploitation harder [12].

LineageOS[4] provides a lightweight, broadly supported AOSP-based system with minimal preinstalled software and extensive device compatibility. It can be deployed in multiple configurations: without Google services, with proprietary Google Play Services and Play Store (Gapps[5]), or with microG[6] as an open-source reimplementation of key Google components. These configurations represent different trade-offs between privacy, compatibility, and reliance on proprietary services.

Finally, iodéOS[7] is a privacy-oriented distribution based on LineageOS that combines a de-Googled baseline with additional privacy features and preconfigured app ecosystem choices, shipping with microG. For completeness, CalyxOS[8] and /e/OS[9] follow a similar privacy-oriented direction (AOSP-based, microG-enabled, and designed for practical daily use). They were considered for the study, but were excluded because they did not receive updates to the current Android version during the measurement period.

The rest of the paper is structured as follows. Section 2 reviews related work concerning network traffic analysis and Android privacy. Section 3 details the experiment design, including the hardware, software, and specific measurement scenarios utilized in our study. Section 4 presents the evaluation results, highlighting data usage across configurations and specific packets analysis. Section 5 discusses the broader implications of these findings, along with study limitations. Finally, Section 6 concludes the paper and outlines potential directions for future research.

2 Related work

Prior work has established that Android privacy risks arise not only from user-installed applications but also, from the operating system and its preinstalled components [5, 6, 4, 3, 11]. Privacy-focused Android distributions have gained

[3] `https://grapheneos.org/`
[4] `https://lineageos.org/`
[5] `https://mindthegapps.com/`
[6] `https://lineage.microg.org/`
[7] `https://iode.tech/iodeos/`
[8] `https://calyxos.org/`
[9] `https://e.foundation/e-os/`

popularity as alternatives to stock vendor firmware, yet they remain comparatively understudied in the measurement literature. Consequently, there is limited empirical understanding of what practical privacy and control benefits different Android distributions provide to users.

To address this gap, a OS-level measurement study compares user privacy of several vendor-customized Android variants with two open-source distributions [5]. It reports that all tested stock/vendor builds transmit substantial data to OS vendors and third parties. Data transmission occurs even when the handset is idle and collection is persistent without an effective opt-out. In contrast, the privacy-focused open-source /e/OS exhibits minimal data transmission. LineageOS still shows substantial communication to Google, because it was evaluated with proprietary Google components (Gapps) present. This distinction is important for studies like ours that evaluate multiple distributions and configurations (including variants with and without integrated Google services). The presence of Google system components can dominate observed traffic and shape the overall privacy profile.

Another work focusing specifically on OEM telemetry further shows that Samsung, Xiaomi, Huawei and Realme make extensive use of long-lived hardware identifiers [6]. They also collect list of installed apps and analytics data, sometimes even in connections that should not require persistent identifiers, enabling straightforward linkage to a user's identity when an OEM account is used.

Related work by the same authors analyzes telemetry from two core Google system applications: Google Messages and Google Dialer, by decrypting and decoding the logging channels that forward data to Google [4]. The study reports that these applications disclose sensitive communication metadata, including when SMS messages and calls are sent or received, with timestamps and for calls duration. Google collects hash derived from message content that can uniquely identify messages and link participants. Other findings show that phone numbers may be transmitted and events are tagged with persistent identifiers such as the Android ID, undermining anonymity and leaving users with no effective opt out. Overall, the results suggest that even basic default communication apps on stock Android cannot be assumed to be trustworthy from a privacy perspective.

Complementing OS-level telemetry measurements, a recent study focuses specifically on what pre-installed Google components persistently store on-device [3]. It shows that Google's role in Android privacy extends beyond outbound traffic. Google Play Services, Play Store and other bundled Google applications receive and store multiple cookies, advertising-related identifiers, and tracking links. Some data persists even after a factory reset and is transmitted when user has not opened Google applications, with no dedicated consent prompt or practical opt-out. The study highlights that Google Android ID as a persistent identifier provisioned early and widely reused in subsequent Google connections. It also documents how additional tokens and cookies can be stored and later transmitted alongside telemetry or analytics events. These findings support that Google components play a central role in Android privacy. They are shaping

device identification and data handling independently of user-installed applications.

An interesting report [11] examines Google's privacy from a broader perspective. It describes data collection as an ecosystem across platforms, applications and advertising services, not as a single feature of Android. It distinguishes "active" data sharing (when users directly use Google products) from "passive" collection that happens in the background. It highlights the risk of Google's tracking and analytics components embedded in many third-party applications and websites. The report argues that these different data sources can be combined to build detailed user profiles. It demonstrates that "pseudonymous" identifiers can still be linked back to user accounts when they appear together in the same workflows.

Prior work consistently indicates that Google is a dominant driver of privacy-relevant data flow on Android. It enables persistent device identification, background telemetry and cross-context linkage between application usage, web activity and advertising infrastructure. At the same time, the literature suggests that meaningful reductions in unsolicited communication are possible when Google components are absent or replaced with open source variants [5]. Alternative configurations and distributions remain less widely deployed, less familiar to typical users, and comparatively under-explored in reproducible measurement studies. This motivates systematic comparisons of stock systems and privacy-oriented alternatives, which we perform in this work to quantify privacy improvements and their trade-offs. While emulator-based setups enable highly reproducible traffic capture [1], we used a physical device to analyze behavior in a real environment, at the cost of reduced control and reproducibility.

3 Experiment design

3.1 Hardware and software

We used a single Google Pixel 6a device (codename *bluejay*) to install and evaluate studied Android distributions. This model was chosen due to its broad compatibility with alternative operating systems. All tested builds base on Android 16 and were obtained in December 2025:

- Stock Android: `BP4A.251205.006`;
- LineageOS: `lineage-23.0-20251206` (`BP2A.250805.005`);
- LineageOS for microG: `lineage-23.0-20251203-microG` (`BP2A.250805.005`);
- GrapheneOS: `2025121200` (`BP4A.251205.006`);
- iodéOS: `7.0-20251129-bluejay` (`BP2A.250805.005`).

Network traffic analysis on Android is challenging due to enforced encryption, certificate pinning, and anti-tamper mechanisms. Traffic was captured using a controlled main-in-the-middle (MITM) gateway with a Linux host acting as Wi-Fi router, redirecting HTTP(S) traffic through an intercepting proxy. QUIC over UDP/443 was blocked to force TCP-based TLS. The proxy CA was installed as

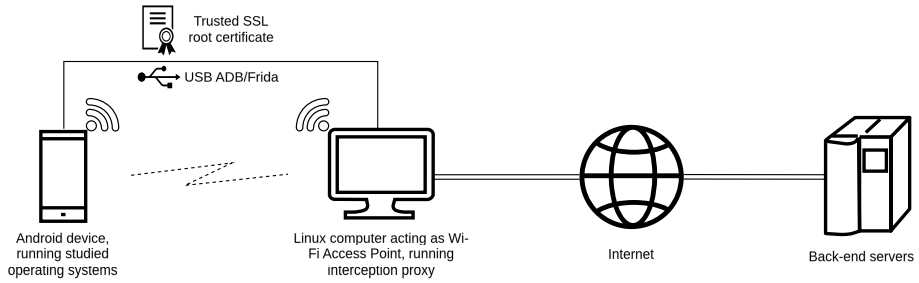

Fig. 1. Measurement setup. Android device running investigated operating systems is configured to access Internet using Wi-Fi. Access point is hosted on a Linux computer, running traffic interception and capturing software. Custom system certificate is installed on the phone using root access. USB connection is used for ADB/Frida communication.

a system-trusted certificate (requiring root). Runtime unpinning via dynamic instrumentation was attempted where certificate pinning blocked interception. Full packet captures were recorded and compared across distributions. We acknowledge that we will not be able to capture and analyze all traffic. Instead we focus on identifying systematic differences between Android distributions and general behavioral patterns. The preview of measurement setup is shown in Fig. 1.

3.2 Measurement scenarios

To enable consistent and comparable measurements across Android distributions, four scenarios were defined. Each scenario specifies the Google account state, consent and permission posture, and which Google components are present. Not all scenarios apply to every operating system due to their characteristics. Scenarios were adapted to each system's capabilities: on GrapheneOS, sandboxed Google Play Services were installed immediately after setup; on systems with microG, the component was enabled during initial setup when prompted, or activated manually immediately afterward. As a general principle, any required configuration step not available during initial setup was performed as soon as possible thereafter.

Scenario A1: Default Setup with Google Account. The device is configured to reflect a typical user path. Google account sign-in is performed during setup (or immediately after) and default recommendations are accepted (including prompted permissions). This scenario is applicable to all systems, except LineageOS without any form of Google services.

Scenario A2: Minimal Setup with Google Account. This scenario extends *Scenario A* by adjusting settings towards privacy while retaining Google account functionality. During and after initial setup, all optional consents and permissions are declined. System and account settings are adjusted to maximize user

privacy. For privacy-focused systems (e.g., those using microG or GrapheneOS), the distinction between Scenario A1 and Scenario A2 is minimal. These platforms are already designed to limit data collection and do not expose the same range of privacy-invasive options present in the factory configuration. Therefore, this we evaluated only scenario A1 for such systems.

Scenario B: Minimal Setup without Google Account. The device is configured with maximum privacy in mind, avoiding Google account sign-in entirely. During initial setup, only strictly required prompts are accepted and optional data-sharing is declined. System settings are adjusted to maximize user privacy. Google components or their functional replacements remain present and enabled to preserve baseline device functionality and application compatibility, but operate without account. This scenario is applicable to the same systems as Scenario A1.

Scenario C: Extremely Minimal Setup without Google Components. This scenario applies only to distributions that do not bundle or enable Google services by default. It is not applicable to the official factory system, which enables Google components by default. If microG is preinstalled, it remains disabled. We did not attempt to disable Google services on stock Android, as doing so significantly degrades or breaks core system and application functionality.

3.3 Measurement tasks

We performed the following measurement tasks sequentially for each scenario. Network traffic was captured using Wireshark[10] for basic analysis.

Task 0: Initial setup. The device was configured according to the target scenario. Network traffic capture began with the device's first Internet connection and continued until 10 minutes after setup completion.

Task 1: System idle. All pending automatic updates were allowed to install and complete. The device was rebooted to begin capture, then kept connected to the monitored network for at least 30 minutes. During this period, the device was unlocked three times and the application list was scrolled through. No applications where opened.

Task 2: Basic applications interactions. Basic preinstalled applications were opened and simple actions (which should not require an Internet connection) where performed:

- Settings: disable haptic feedback; toggle dark mode; check per-application battery usage;
- Clock: set an alarm;

[10] `https://www.wireshark.org/`

– Files: create a folder;
– Contacts: create a local (on-device) contact.

Beyond direct task comparisons, we choose specific packets for detailed analysis to compare data transmission practices across different system configurations. It helped to characterize what data is sent and how. This analysis required TLS decryption via the intercepting proxy and, in some cases, runtime certificate unpinning with Frida using scripts from HTTP Toolkit.[11] Although Frida-based unpinning failed in most attempts, it was useful in some cases. Where needed, we used publicly available Protocol Buffer definitions from microG's source code[12] to decode payloads.

In addition to network-level observations, Google Takeout[13] was used to download data collected by Google for test accounts. It allowed us to inspect and compare server-side data associated with running scenarios involving different implementations of Google services (Stock Android, sandboxed Google Play, microG).

4 Results

4.1 Data usage across operating systems and scenarios

Total network data usage across all operating systems and applicable scenarios is presented in Figures 2–4. To keep the results comparable and avoid dominance by bulk transfers, we excluded traffic to domains used for application and system update payload delivery (large binary downloads). However, this exclusion was not perfect, and some residual update-related traffic remains in the dataset. Generally, we expect lower network traffic in configurations with fewer Google services present.

Figure 2 illustrates network activity during initial device setup. As expected, stock Android shows the highest data usage (71-85 MB). The modest reductions between privacy-adjusted scenarios indicate that configuration choices provide little control over automatic updates and background activity. Configurations incorporating more Google services generate higher traffic, while de-Googled baselines demonstrate substantially reduced data exchange. LineageOS with official Google Play Services generates substantially less traffic (16-21 MB) than stock Android due to its minimal base system. However it remains considerably higher than configurations without proprietary Google services, with the microG variant and Google-free installation achieving notable further reductions (0-3 MB). For GrapheneOS, Scenario B demonstrates that even when sandboxed Google Play Services are installed and granted network and background permissions, they remain largely inactive when not deliberately invoked. This contrasts with the persistent background activity characteristic of integrated GMS deployments.

[11] `https://github.com/httptoolkit/frida-interception-and-unpinning`
[12] `https://github.com/microg/GmsCore/`
[13] `https://takeout.google.com/`

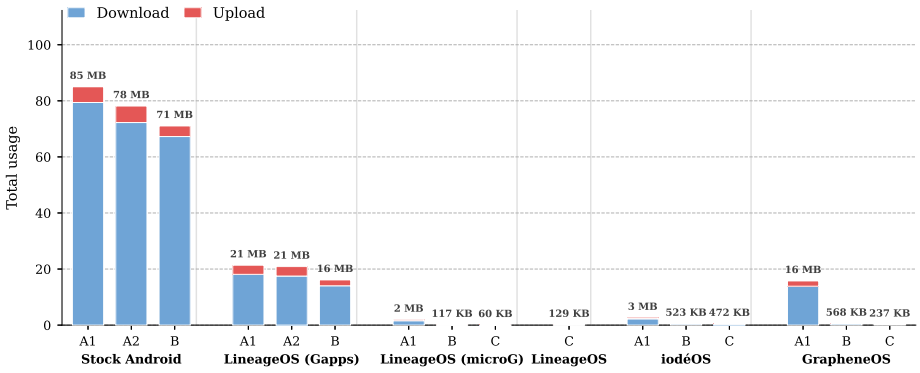

Fig. 2. *Task 0: Initial setup* data usage. Scenario applicability varies by system (e.g., Scenario A1 requires Google services, which are absent on LineageOS).

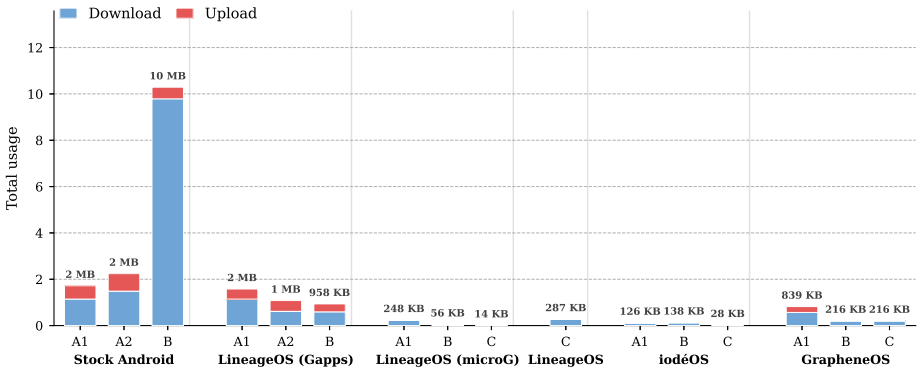

Fig. 3. *Task 1: System idle* data usage. Scenario applicability varies by system (e.g., Scenario A2 would not significantly differ from A1 on GrapheneOS, because stock configuration is already privacy focused).

Network activity observed during system idle following a reboot is presented in Figure 3. Stock Android again shows the highest data usage, though Scenario B exhibits unexpectedly elevated traffic compared to both Scenario A2 and Scenario A. This anomaly results from substantial background communication with *www.gstatic.com*, likely triggered by automatic updates or other background activity during the measurement time.

LineageOS in Scenario C similarly deviates from anticipated patterns, generating more idle traffic than its microG variant. GrapheneOS exhibits nearly identical traffic in Scenarios B and C, confirming that sandboxed Google Play Services generate minimal background activity when not actively in use.

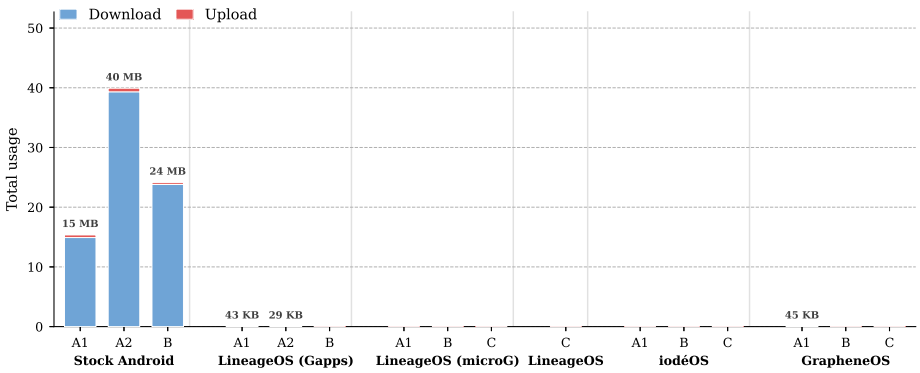

Fig. 4. *Task 2: Basic applications interactions* data usage. Scenario applicability varies by system (e.g., Scenario C is not applicable to Stock Android, which bundles Google services by default.

Figure 4 shows network activity during basic application interactions that should not require Internet connectivity. Stock Android dominates with substantial data usage mostly by uncontrolled updates. All alternative distributions demonstrate significantly lower traffic, with de-Googled configurations remaining effectively silent. Notably, whenever official Google Play Services were present and a Google account logged in, at least one request to Google infrastructure occurred.

4.2 Domain distribution and Google dependencies

Figure 5 presents the domain-level distribution of network traffic for Scenarios A and B. These scenarios were selected as the most realistic configuration,s incorporating essential Google functionality while differing in account linkage and privacy-related settings. Overall traffic is dominated by large content transfers, as systems download necessary resources and updates. Stock Android contacts the widest range of Google domains, distributing traffic across numerous end-

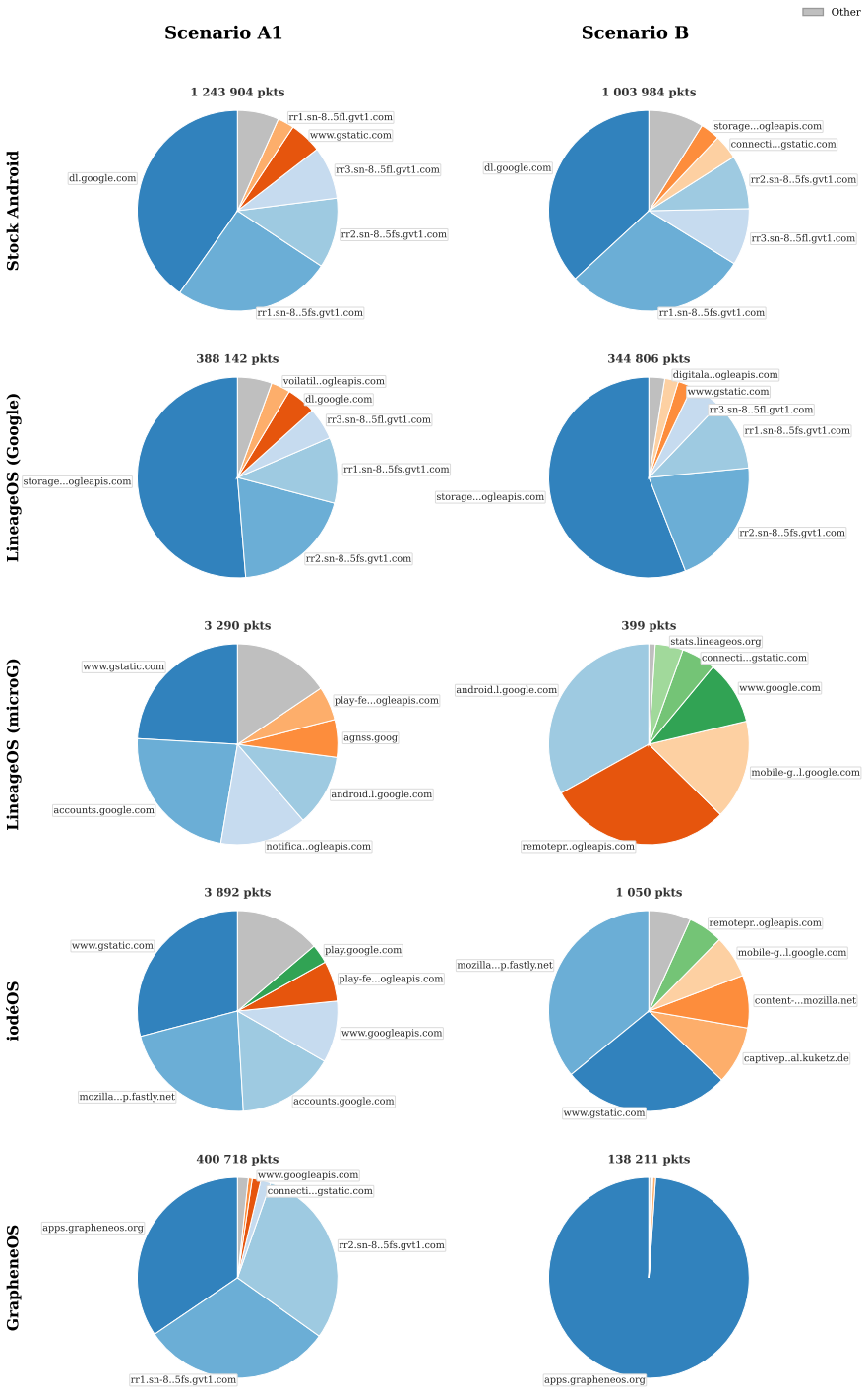

Fig. 5. Domain-level distribution of network packets (pkts) for scenarios A (with Google account) and B (without Google account, but with Google Services present), aggregated over all tasks. Each pie shows the top domains by packet count for a given operating system and scenario.

points. Alternative systems show reduced Google communication, although patterns vary by design. GrapheneOS generates more requests than microG-based systems, as it downloads Google services on demand from its own application store. IodéOS and LineageOS with microG contact fewer Google servers overall, as microG minimizes connectivity while still enabling basic functionality. IodéOS additionally relies on some Mozilla infrastructure not present in other systems.

Table 1 shows the infrastructure dependencies for core system functions. LineageOS variants remain closest to stock Android, relying entirely on Google servers for connectivity checks, remote provisioning, and location assistance. IodéOS adopts an intermediate position, using independent infrastructure for connectivity checks while still depending on Google for remote provisioning and location services. GrapheneOS demonstrates the strongest de-Googling, substituting all three functions with its own or independently operated infrastructure. Both iodéOS and GrapheneOS expose these mechanisms through explicit system settings toggles, though GrapheneOS offers more granular control: users may choose between its independent proxy infrastructure and official Google servers for each function, with default configured to avoid Google connections entirely.

Table 1. Domains observed for connectivity check, remote provisioning, and location assistance.

System	Connectivity check	Remote provisioning	Location assistance
Stock Android	connectivitycheck.gstatic.com	–	supl.google.com agnss.goog
LineageOS	connectivitycheck.gstatic.com	remoteprovisioning.googleapis.com	supl.google.com agnss.goog
iodeOS	captiveportal.kuketz.de	remoteprovisioning.googleapis.com	supl.google.com
GrapheneOS	connectivitycheck.grapheneos.network	remoteprovisioning.grapheneos.org	gs-loc.apple.grapheneos.org

4.3 Additional findings

Beyond aggregate traffic measurements, we examined specific data transmissions revealed through TLS interception and payload analysis. A subset of these observations originates from earlier experiments performed on Android 14. They could not be reconfirmed on Android 16, as the Frida-based method we employed became unsuccessful, though we expect the general patterns to persist. GrapheneOS blocked all our instrumentation attempts, so we cannot verify what data these requests contained. However, the substantially lower request volume suggests that sandboxed Google Play Services likely transmit less complete or less authentic information.

Android's checkin mechanism transmits device identifiers to Google infrastructure. On stock systems, this includes the real IMEI and hardware serial number. In contrast, microG employs a privacy-preserving approach by generating randomized, rotating identifiers including fake IMEIs (consistently prefixed with `35503104`), MAC addresses (with OUI `b4:07:f9`), and a synthetic serial number displayed in microG settings. The Android ID field in microG requests

is frequently empty or zeroed. However, microG still transmits authentic device characteristics including model, Android version, and timezone, enabling functional compatibility while obscuring persistent hardware identifiers.

We observed requests to *app-measurement.com* on systems with Google services present, primarily on stock Android. These transmissions contained application package names, installation sources (Google Play, Aurora Store, manual), versions and system theme preference. Information about installed applications was transmitted even for the ones that were never launched.

The Google Advertising ID (ADID) was observed in multiple requests on stock Android, formatted as a UUID. This value remained constant across different requests, confirming its persistence as a cross-application tracking mechanism. After a manual reset in system settings, the ADID was transmitted as all zeros, indicating the identifier was actually cleared rather than immediately regenerated.

When enabled, microG performs network geolocation without Google infrastructure by querying a selected provider, in our case *api.positon.xyz*. The request transmits data about nearby cell towers and returns approximate coordinates. Location fixes were obtained almost immediately after enabling this feature, compared to significantly slower location acquisition prior to activation.

We compared Google Play Store with Aurora Store, a popular application source in alternative Android distributions, by downloading the same application from each platform. Both stores retrieved identical packages from the same Google CDN infrastructure. However, the Play Store generated 106 requests, while Aurora Store produced 93, omitting Google telemetry connections, but adding lookups to Exodus Privacy and Plexus databases for tracker analysis and compatibility ratings. Aurora achieves functional equivalence by mimicking Play Store user agents and headers, using shared anonymous credentials rather than personal account.

4.4 Google Takeout

We submitted Google Takeout requests for accounts we used, and analyzed the Android device data. Four distinct Google service configurations where examined: Stock Android as a baseline, LineageOS with Gapps as an alternative system incorporating Google services, LineageOS with microG as an open-source reimplementation (IodéOS includes an identical microG version), and GrapheneOS as a representative of sandboxed approach. For every logged in system, a corresponding entry appeared in `devices.json`, including device model, OS version, country, language, registration timestamp, and last activity time. However, the depth of telemetry varied considerably across configurations, as summarized in Table 2.

Both systems using official Google services reported a large list of installed applications. GrapheneOS reported only four preinstalled applications, two of which were GrapheneOS specific. In contrast, microG did not create any application-related entry. Additionally, Stock Android and LineageOS with GMS both transmitted authentic hardware identifiers including IMEI and serial number. Graphe-

neOS submitted a falsified serial number and omitted IMEI entirely. Again, microG did not create additional device-related entry, so these identifiers were not present. A Pixel specific telemetry directory, present only for Stock Android and LineageOS with GMS, contained extensive hardware monitoring data including power measurements, Wi-Fi signal logs, battery capacity readings, thermal samples, and battery cycle entries. LineageOS reported less granular telemetry than Stock Android, indicating that while basic Pixel telemetry infrastructure remains functional, the depth of hardware monitoring is reduced compared to the proprietary stock implementation.

Table 2. Comparison of Google Takeout data across Android configurations.

System	Device identifiers	Installed apps	Pixel telemetry
Stock Android	IMEI, serial number	Extensive list	Full (power, Wi-Fi, battery, thermal)
LineageOS (Gapps)	IMEI, serial number	Extensive list	Reduced
LineageOS (microG)	–	–	–
GrapheneOS	Falsified serial number	Minimal (4 apps)	–

5 Discussion

Our measurements reveal a clear hierarchy in how different Android configurations handle user data and privacy. The most fundamental finding is that adjusting privacy settings on stock Android, while beneficial, cannot provide the same level of protection as switching to an alternative distribution. On systems running proprietary Google services, privacy-adjusted scenarios showed only modest reduction in data transmission compared to the default setup. This is due to the underlying architecture which tightly integrates Google components into core of OS and maintains persistent background activity. As a result, substantial data flows to Google infrastructure continue regardless of user-facing privacy settings.

Alternative distributions shift the privacy posture by making Google communication user-initiated rather than ambient. All privacy-focused systems generated orders of magnitude less network traffic with Google. Some of them utilize their own or third-party privacy-focused infrastructure to replace Google services. LineageOS with GMS was an exception, while more minimal, by having official Google services preinstalled, it does not differ much from stock Android in terms of privacy. Among other configurations enabling basic Google functionality, microG represents the most privacy-preserving approach, though this comes with potential security trade-offs (signature spoofing, privileged install). Our measurements show that microG transmits the least data to Google infrastructure while maintaining basic service compatibility. It achieves this through identifier randomization rather than exposing authentic hardware identifiers, similar to GrapheneOS. Our observations illustrate the fundamental difference: microG optimizes for privacy, whereas GrapheneOS prioritizes security, but still providing strong privacy, substantially improved over stock systems.

Alternative distributions also improve user control through minimal base systems and enhanced permission granularity. All tested alternatives ship without forced user agreements during initial setup, unlike stock Android, which requires accepting mandatory terms including data collection. A notable privacy-enhancing feature present in all tested systems, which is absent in stock Android, is per-application network permission control, allowing users to restrict Internet access for specific applications. Aurora Store demonstrates that downloading applications from official repositories remains viable without full Google infrastructure interaction, retrieving identical packages from the same source while generating fewer requests. Updates in alternative systems are typically smaller, more transparent, and user-controlled rather than automatic and opaque.

Several limitations qualify these findings. Our measurements capture specific, time-bounded scenarios rather than universal behavior. Traffic patterns likely vary with extended use and evolving service configurations. The TLS interception approach, while revealing substantial information, remains incomplete due to certificate pinning, anti-tamper mechanisms, and ongoing protocol evolution. GrapheneOS's resistance to our instrumentation, while security-advantageous, means we cannot closely verify what data sandboxed services transmit. These constraints indicate that our findings reflect general trends and architectural differences rather than definitive or absolute measurements.

Based on the network traffic we analyzed, Google may potentially distinguish between stock Android and alternative distributions through device characteristics transmitted during service interactions, raising the possibility of selective enforcement or access restrictions. Ultimately, the choice between privacy-focused alternative operating systems depends on individual threat models and functional requirements, but all of them represent meaningful improvements in user control and privacy over stock Android.

6 Conclusion

In this work, we compared network behavior across multiple Android distributions, measuring data transmission during setup, idle periods, and basic application usage. We evaluated Stock Android, LineageOS variants, iodéOS, and GrapheneOS to quantify how different approaches to Google services affect user privacy. Our findings show that alternative distributions reduce unsolicited data flows by orders of magnitude compared to stock systems. These systems enhance privacy by reducing data exposure, limiting or randomizing identifiers, and using independent infrastructure. Network traffic and Google Takeout analysis revealed that stock Android transmits persistent hardware identifiers and various telemetry data, whereas alternative systems either do not transmit such data or significantly limit its scope.

Future work could extend measurements to longer usage periods and repeated trials to distinguish systematic differences from transient variation. More realistic tasks incorporating diverse applications would better approximate everyday use. As Android, Google services and their alternatives continue to evolve, re-

peating this study would help determinate whether future privacy improvements meaningfully alter the patterns we observed.

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